National University of Computer and Emerging Sciences

Lahore Campus

Assignment-I

Information Security (CS3002)

Total Marks: 75

Total Questions: 5

I. **Instructions**: Solve each question on paper, showing all steps clearly. Ensure clarity and conciseness in your explanations.
II. Assignments must be received before the deadline. Please do the work by yourself, this is an individual assignment.
III. Plagiarism cases will be dealt with strictly.
IV. Read questions and marks distribution carefully and write precise answers, avoiding wordy stories.
V. Make assumptions where needed but state them clearly in your answer.
VI. For *Monoalphabetic Substitution Cipher Q3* the answer will vary as it depends on how you guess the pattern.

VII. BEST OF LUCK!
VIII. Make sure you clearly identify your name, roll number and section.

Questions

1. **Caesar Cipher (5 + 5)**

Q1: Decrypt the following ciphertext, assuming Caesar's cipher with shift of 3:

Ciphertext: "**WKLV ZRUOG LV WHPSRUDUB**"

Show your work: Include each shift for the individual characters and explain how the encryption works.

Q2: Encrypt your Instructor Name by using Caesars's Cipher. Key should be the default value.

2. **Monoalphabetic Substitution Cipher (4 + 4 + 7)**

Q1: Encrypt the following message using the substitution cipher where each letter is mapped as follows:

Plaintext: "**INFORMATION SECURITY**"

Substitution Key: A → Q, B → W, C → E, D → R, E → T, F → Y, G → U, H → I, I → O, J → P, K → A, L → S, M → D, N → F, O → G, P → H, Q → J, R → K, S → L, T → Z, U → X, V → C, W → V, X → B, Y → N, Z → M.
Show your work: Demonstrate the substitution process for each letter.

Q2: You receive a ciphertext message: **"KOLT XH"**. Using the above substitution key, decrypt it to find the original plaintext.

Q3: Decrypt the following text by using Frequency analysis
IEAWEOEATEUNERTAARTOHNLEISTEDYSCFGSEMTIHEOKJNTEHIS
Note: Clearly state Frequency table and explain how you solved it and what are the challenges you faced while doing cryptanalysis.

3. **Vigenère Cipher (5 + 5)**

Q1: This problem explores the use of a one-time pad version of the Vigenère cipher. In this scheme, the key is a stream of random numbers between 0 and 26. For example, if the key is 3 19 5 ..., then the first letter of plaintext is encrypted with a shift of 3 letters, the second with a shift of 19 letters, the third with a shift of 5 letters, and so on.

a. Encrypt the plaintext "**wehaveexamsfromsaturday**" with the key stream:

9 0 1 7 23 15 21 14 11 11 2 8 9

b. Using the ciphertext produced in Part (a), find a **key** so that the cipher text decrypts to the plaintext "**youhavegotthiskeepgoing**".

4. **Rail Fence Cipher (5 + 5)**

Q1: Use a rail fence cipher with depth 3 to encrypt the following plaintext:
"EARNYOURSUCCESS".
Show the Work: Write out the zigzag pattern and read the encrypted message.

Q2: Given the ciphertext **"TSNRHSIETEYYIAGMIVESSNSA"**, decrypt it using a rail fence (try to find out the depth level by yourself)
Explain: Illustrate the reverse process of zigzagging and retrieving each row.

5. **Columnar Transposition Cipher (5 + 5 + 5)**

Q1: Encrypt the plaintext **"ASSIGNMENT IS DUE ON THURSDAY"** using a columnar transposition with key order [3, 1, 4, 2].
Steps: Organize the letters in a grid and show the column-wise reading order.

Q2: Decrypt the ciphertext **"SOC HSSE TIPR ITET"** using the key order [3, 2, 4, 1].

Explanation: Lay out the ciphertext in the grid, and demonstrate how to read each column in the key's sequence to reveal the plaintext.

Q3: By using columnar transposition, Perform encryption on the following text:

Plaintext = '**HIDDENTRUTHSBEYONDCOMPARE**'

Key = **UNCLOAK**

6. **RSA & Difie Hellman (7.5 + 7.5)**

Q1: In the RSA algorithm, given that the **p = 13** and **q=17**. You need to calculate the following items:

1. The value of n.
2. The totient φ(n).
3. If the public exponent **e=35** private key exponent d.
4. Perform encryption and decryption.

encrypt and decrypt the message m=10

Public key(e,n)

Private key(d,n)

Encryption = C = P^e mod n

Decryption = P = C^d mod n

Q2: Suppose Alice and Bob wish to perform Difie Hellman key exchange. They agree on prime number p = 11 and generator g = 6. Alice chooses a secret 5, while Bob chooses 8. An attacker Mallory is listening to all of their communication. He chooses his own secret 4.

Show the calculation that proves that Mallory can perform a man-in-the-middle attack against Alice and Bob's key exchange.

Submission: Ensure that each step is presented clearly, with the process fully outlined to demonstrate your understanding of each cipher. Submission will be on gcr and you have to submit hard copy of your assignments to your instructor as well.